

QCon

全球软件开发大会

从 LLM 到 Omni: vLLM-Omni 的全模态推理架构演进

朱江云

vLLM-Omni Committer



极客邦科技 2026 年会议规划

促进软件开发及相关领域知识与创新的传播



参会咨询

🕒 6月 📍 上海

👥 1000人

AiCon

全球人工智能开发与应用大会

会议时间：6月26-27日

- AI Infra 系统工程
- 多 Agent 协作与实践
- 多模态融合
- 模型训练与推理创新
- 数据平台与特征服务

🕒 8月 📍 深圳

👥 1000人

AiCon

全球人工智能开发与应用大会

会议时间：8月21-22日

- Agentic AI
- 轻量化与高效推理
- 多模态应用
- AI + IoT 场景实践
- AI 工业化落地

🕒 10月 📍 上海

👥 1200人

QCon

全球软件开发大会

会议时间：10月22-24日

- AI Agent
- Vibe Coding
- 智能可观测
- 推理基建
- 模型攻防
- AI x 创造力

🕒 12月 📍 北京

👥 1000人

AiCon

全球人工智能开发与应用大会

会议时间：12月18-19日

- 大模型架构创新
- 多模态 AI 产业融合
- 具身智能
- AI for Science
- 大模型安全

About us



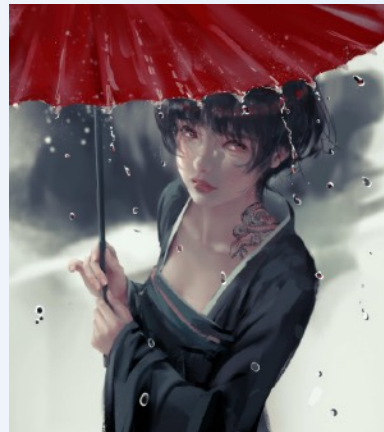
Roger Wang
vLLM&vLLM-Omni
Committer



Hongsheng Liu
vLLM-Omni
Committer



Han Gao
vLLM-Omni
Committer



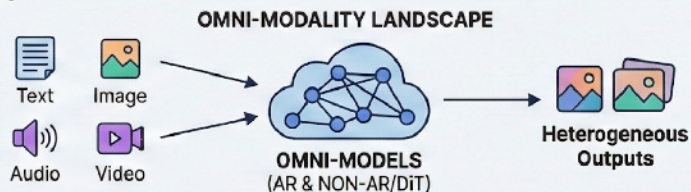
Jiangyun Zhu
vLLM&vLLM-Omni
Committer

TRADITIONAL SERVING (Text-only AR)



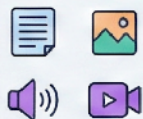
THE SHIFT TO OMNI-MODALITY

EVOLVING
INFRASTRUCTURE

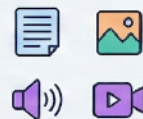


vLLM-OMNI ARCHITECTURE: RE-IMAGINED HETEROGENEOUS PIPELINE

MULTIMODAL INPUTS



RICH MEDIA OUTPUTS



KEY FEATURES & BENEFITS

SIMPLICITY



Seamless Integration
(OpenAI API, Hugging Face Models).
Easy to Use if you know vLLM.

FLEXIBILITY



OmniStage Abstraction
Supports various Omni-Models
(e.g., Qwen-Omni, Qwen-Image).

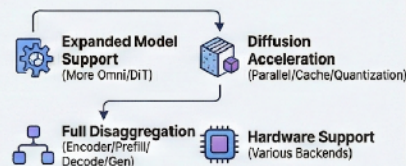
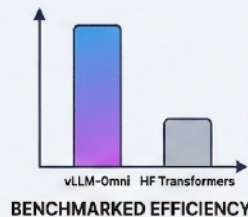
PERFORMANCE

PIPELINED STAGE EXECUTION



Overlapping Computation for High Throughput
Efficient Memory Management.

PROVEN EFFICIENCY & FUTURE ROADMAP

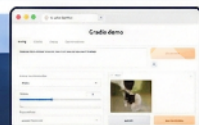


ROADMAP & DEVELOPMENT

GET STARTED & JOIN THE COMMUNITY

Installation & Examples: github.com/vllm-project/vllm-omni
Documentation: docs.vllm.ai/projects/vllm-omni/en/latest

Slack: #sig-omni at slack.vllm.ai
Weekly Meeting: Wednesday 19:30 PDT



Our Goal

Build the **fastest** and
easiest-to-use open-source

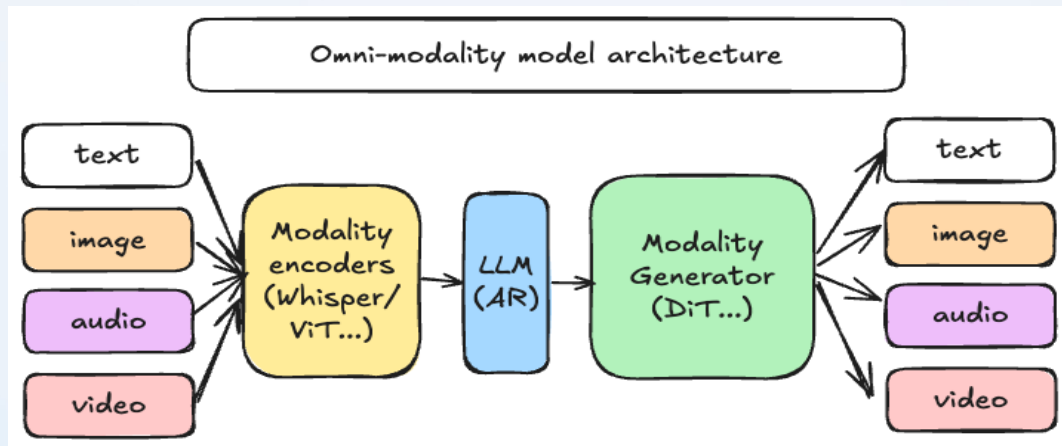
Omni-Modality model inference &
serving engine

Omni-Modality models

Omni-modality: Text, image, video, and audio data processing

Non-autoregressive Architectures: extend the AR support of vLLM to Diffusion Transformers (DiT) and other parallel generation models

Heterogeneous outputs: from traditional text generation to multimodal outputs



vLLM-mni API (1): **Omni** class

A Python interface for offline batched inference for Qwen3-Omni/Qwen-Image

```
from vllm\_omni import Omni
```

Example prompts.

```
inputs = {"prompt": prompt,  
         "multi_modal_data": {"video":  
                               video_frames, "audio": audio_data},}
```

Create an omni with HF model name.

```
omni = Omni(model="Qwen/Qwen3-  
Omni-30B-A3B-Instruct")
```

Generate texts and audio from the multi-modality inputs.

```
outputs = omni.generate(inputs)
```

```
from vllm\_omni import Omni
```

Example prompts.

```
inputs = "A cup of coffee on the table"
```

Create an omni with HF model name.

```
omni = Omni(model="Qwen/Qwen-  
Image-2512")
```

Generate image from multi-modality inputs.

```
outputs = omni.generate(inputs)
```

vLLM-Omni API (2): OpenAI-compatible server

A FastAPI-based server for online serving for Qwen3-Omni

Server

```
$ vllm serve Qwen/Qwen3-Omni-30B-A3B-Instruct --omni --port 8091
```

Client

```
$ curl -sS -X POST http://localhost:8091/v1/chat/completions\  
-H "Content-Type: application/json" \  
-d '{  
  "model": "Qwen/Qwen3-Omni-30B-A3B-Instruct",  
  "messages": "Why is this video funny? "  
  "sampling_params_list": $sampling_params_list,  
}'
```


vLLM-Omni API (2): OpenAI-compatible server

A FastAPI-based server for online serving for Qwen-Image

Server

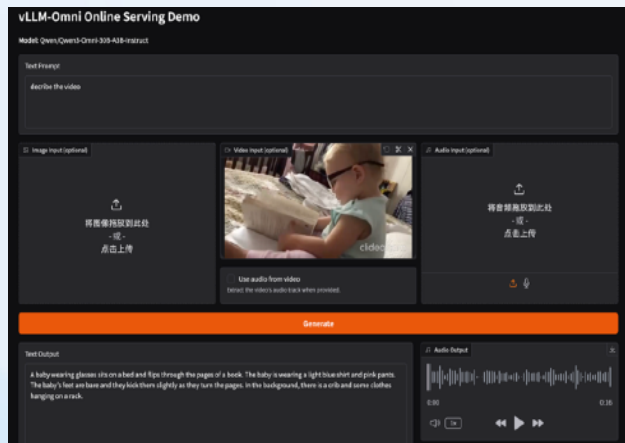
```
$ vllm serve Qwen/Qwen-Image-2512 --omni --port 8091
```

Client

```
$ curl -X POST http://localhost:8091/v1/images/generations \
-H "Content-Type: application/json" \
-d '{
  "prompt": "a dragon laying over the spine of the Green Mountains of
Vermont",
  "size": "1024x1024",
  "seed": 42
}' | jq -r '.data[0].b64_json' | base64 -d > dragon.png
```

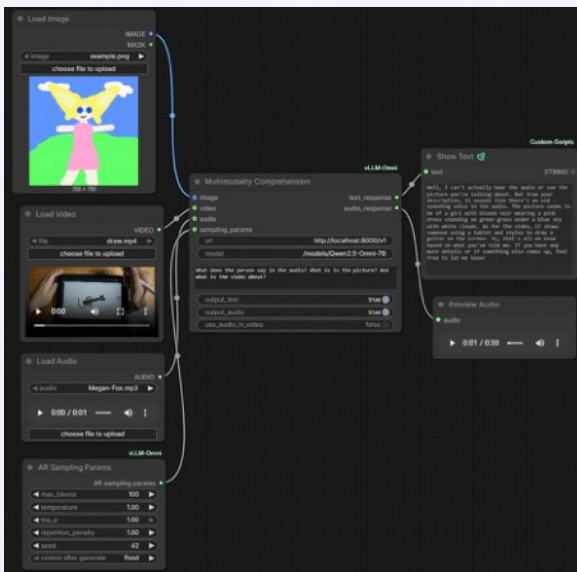
vLLM-Omni serving demo

A Gradio demo for Qwen3-Omni online serving



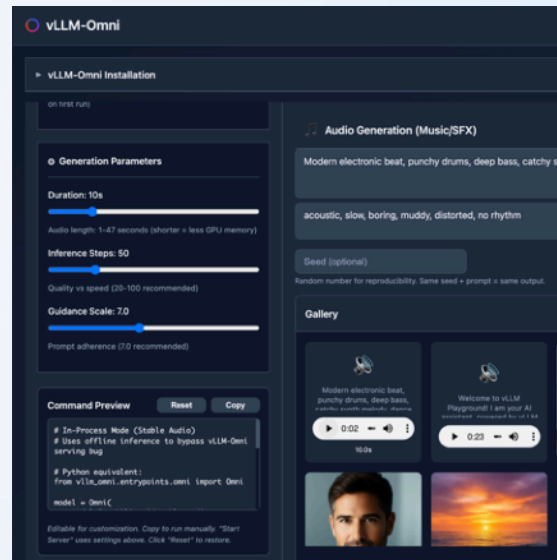
https://docs.vllm.ai/projects/vllm-omni/en/latest/user_guide/examples/online_serving/qwen3_omni/

ComfyUI Integration for low-code development



<https://docs.vllm.ai/projects/vllm-omni/en/latest/features/comfyui/>

vllm-playground: A web interface for interacting with vLLM servers



<https://github.com/micytao/vllm-playground>

Broad Model Support

vLLM-Omni supports 30+ popular omni and diffusion model architectures(growing rapidly)



Qwen @Alibaba_Qwen · Jan 2

Big thanks to @vllm_project for day-zero support of Qwen-Image-2512! 🎉

vLLM @vllm_project · Dec 31, 2025

Congrats to @Alibaba_Qwen on the release of Qwen-Image-2512! 🎉
We are thrilled to announce Day-0 support in vLLM-Omni. You can now serve this SOTA open-source image model with our optimized pipelined architecture immediately.



Qwen-Omni
Qwen-Image



BAGEL



LongCat



Z-Image



Tencent
Hunyuan



SD3



Wan



MiMo



Flux



StepFun



GLM

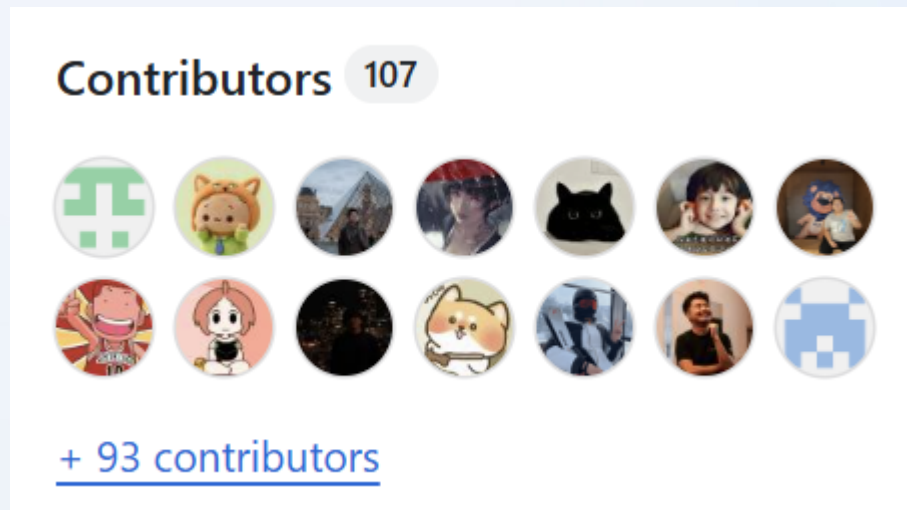


Ovis-Image



Contributors

Thanks to all the **contributors** who raised issues, participated in discussions, and submitted PRs!



...

vLLM Github Repo

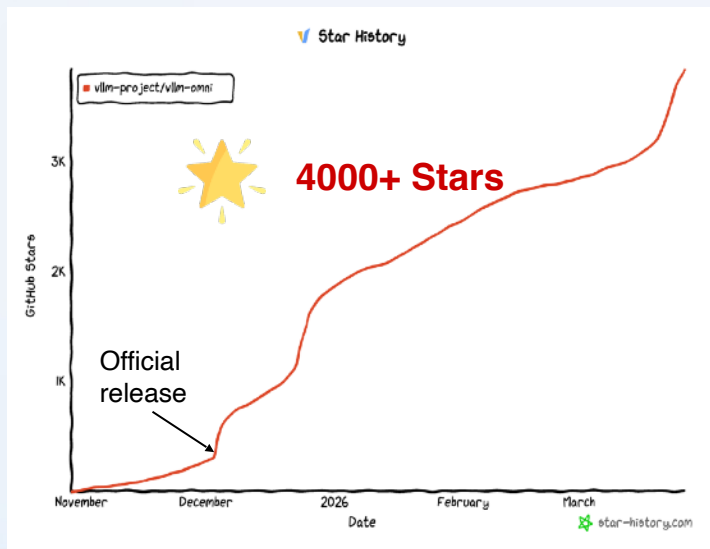


<https://github.com/vllm-project/vllm-omni>



```
$ uv pip install vllm --torch-backend=auto
```

```
$ uv pip install vllm-omni
```



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InfoQ 极客传媒

vLLM-Omni System Walkthrough

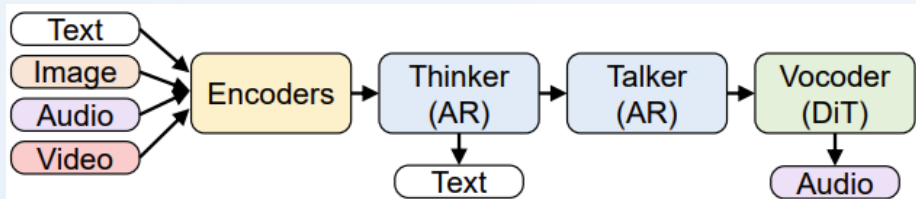
- vLLM-Omni Team

Goal of the walkthrough

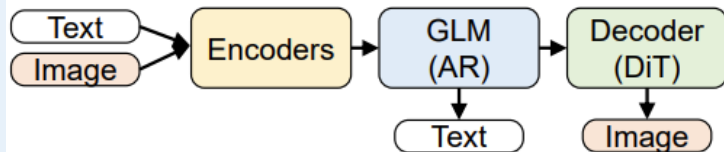
1. Understand how vLLM-Omni processes a **multi-modal** request and generates its **multi-modal** outputs.
2. Learn where to modify if you would like to make a specific modification/contribution.

Multi-modality models

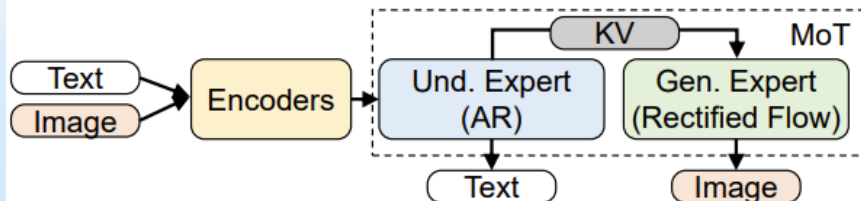
Yin, Peiqi, et al. "vLLM-Omni: Fully Disaggregated Serving for Any-to-Any Multimodal Models." arXiv preprint arXiv:2602.02204 (2026).



(a) Thinker-Talker (Qwen2.5-Omni)



(b) AR + DiT (GLM-Image)



(c) AR + Specialized Generator (BAGEL)

Backbone: (multi) AR + DiT

Models: Qwen-Omni/Ming-Omni

Tasks: any-to-any

Backbone: AR + DiT

Models: Qwen-Image/GLM-Image

Tasks: t2i, t2v, i2i...

Backbone: AR + Spec. Gen.

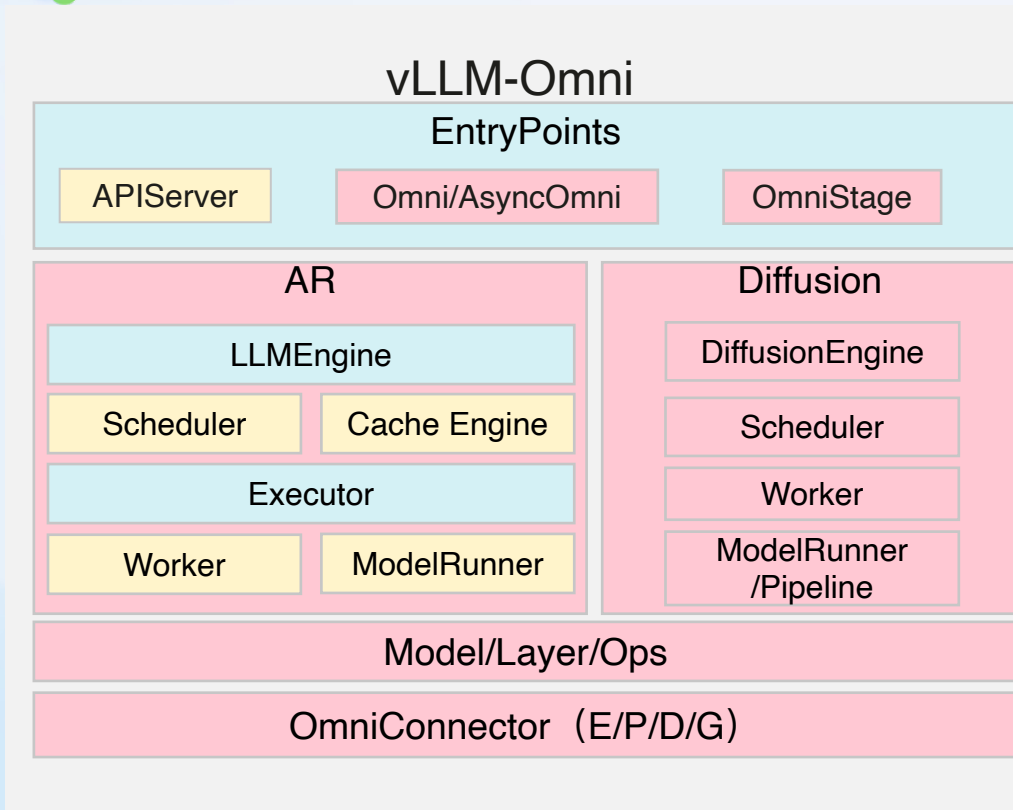
Models: BAGEL, Hunyuan Image 3.0

Tasks: t2i, i2i, i2t...

Multi-modality models: AR/DiT comparison

	AR	DiT
Use cases	Text generation	Multi-modelity generation
Generation process	Token-by-token KV Cache based	Diffusion step
Bottleneck	Prefill: compute bound Decode: memory bound	Compute bound
Seq length	varied	Fixed
Attention Mask	causal mask	Full mask
parallelism	TP/DP/EP/PP/CP/SP	TP/EP/USP/CFG

Main architecture of vLLM-Omni



imported

modified

new

Main component:

- **Entrypoints:** offline/online serving, OmniStage abstraction for model stages(AR/ DiT)
- **AR module:** inherited from vLLM(CB/PA/ Prefix Cache...) and adapted to the Omni-modality model
- **Diffusion module:** implemented natively and optimized by acceleration components
- **Model/Layer/ops:** parallelism, quantization, attention...
- **OmniConnector:** natively supports E/P/D/ G disaggregation



Hardware Plugin System

PR #774

plugin entry

hardware implementation

vLLM-Omni

EntryPoints

AR

LLMEngine

Scheduler

Cache Engine

Executor

Worker
(GPU/NPU/XPU)

ModelRunner
(GPU/NPU/XPU)

Diffusion

DiffusionEngine

Scheduler

Worker

ModelRunner/
Pipeline

Hardware Platform

current_omni_platform

(DiT/ViT)
Attention Selector

Worker Selector

Model/Layer/Ops

RotaryEmbedding

AdaLayerNorm

DiffusionAttentionBackend

Hardware-optimized
kernel

FA

Sag
e

SDPA

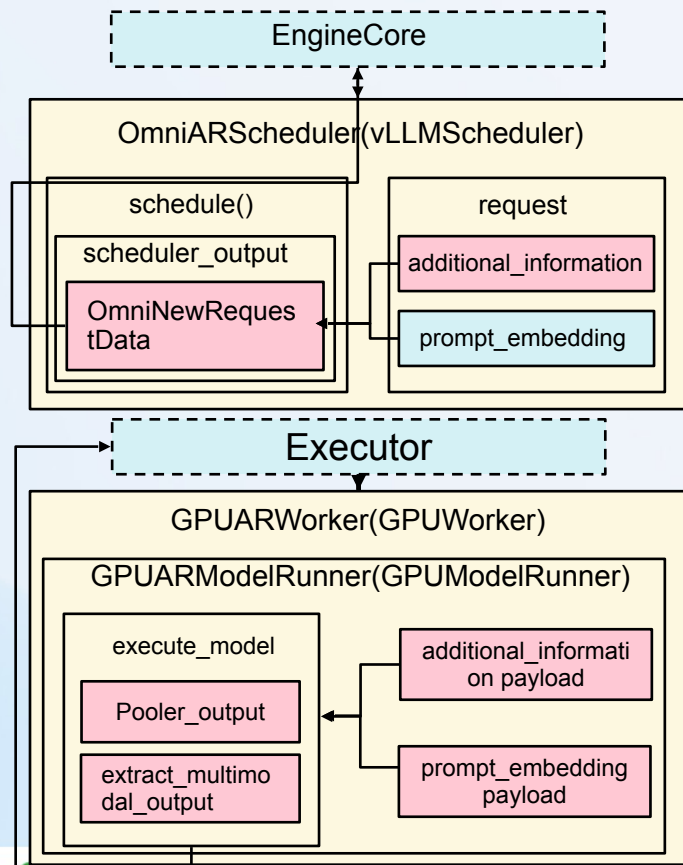
Multi-Hardware Coverage:
CUDA/ROCm/NPU/XPU

Composable Inheritance:
Reuses vLLM's native platform capabilities(for AR) via multiple inheritance while adding Omni-specific interfaces.

Unified Abstract Interface:
current_omni_platform

Op Dispatch:
Attention backends and custom ops are dispatched per-platform with hardware-optimized implementations.

AutoRegressive(AR) Module Design



imported

modified

new

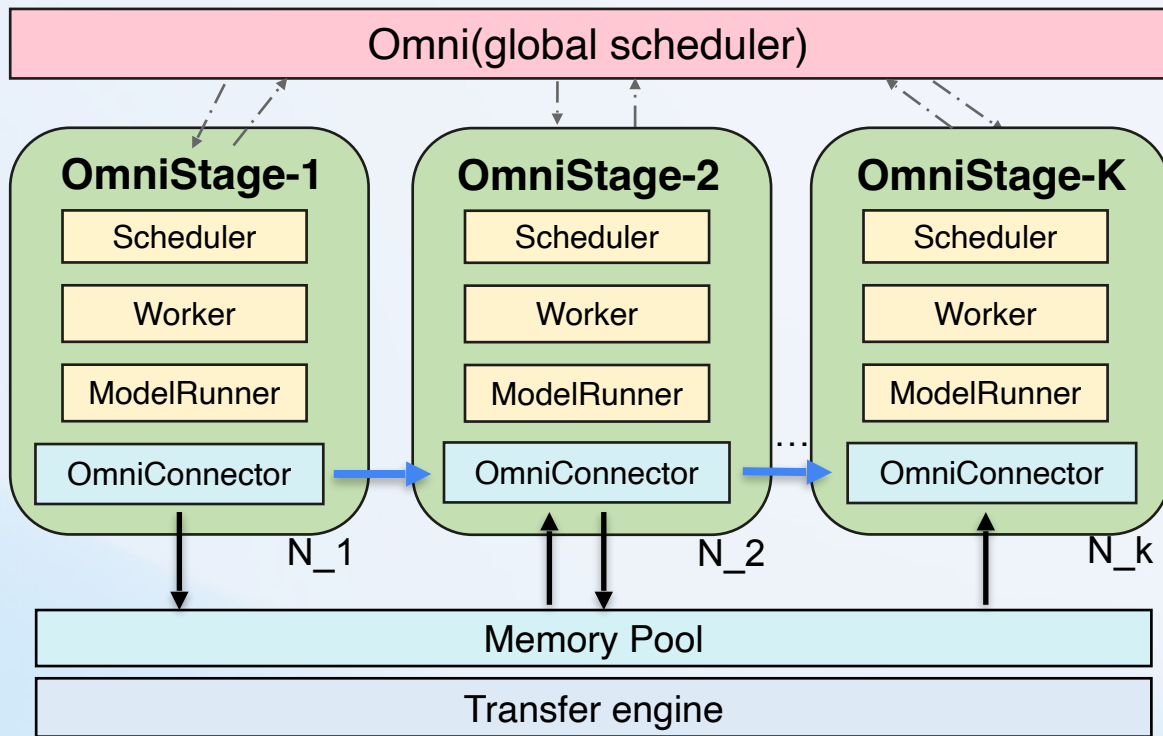
1. Handles autoregressive generation stages:

- Text generation
- Hidden embeddings
- multimodal latent tokens (e.g.: Talker in Qwen3-Omni)

2. Extends vLLM's core to support:

- **Multimodal inputs/outputs:** Processing images, videos, and audio alongside text
- **Prompt Embedding:** Passing pre-computed prompt embeddings between pipeline stages via serialized payloads
- **Additional information:** Carrying per-request metadata (tensors, lists) through the pipeline

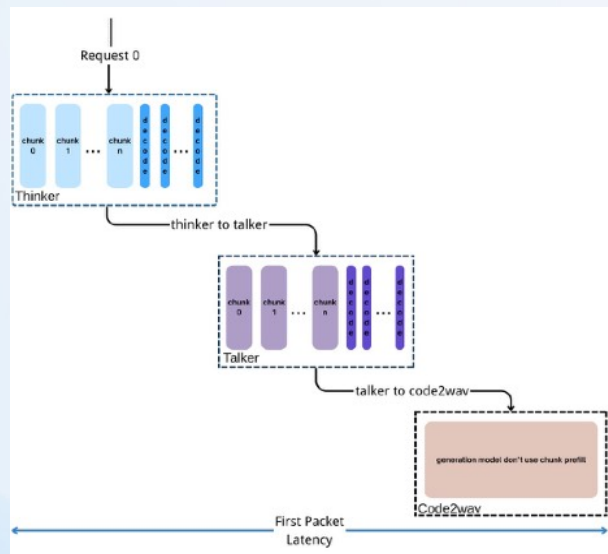
Natively Disaggregated Serving



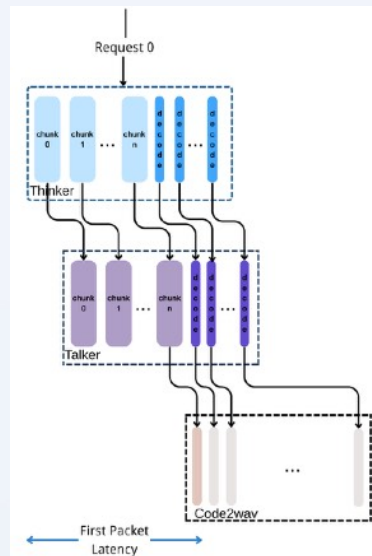
- **Standardized Unified Abstraction:** A generalized interface that handles heterogeneous data (Text, Image, Audio).
- **Control & Data Plane Decoupling:** Metadata travels via lightweight control signals, while heavy payloads are offloaded to high-performance data planes.
- **Hybrid Backend Support:** Native support for Shared Memory (SHM) and Mooncake for distributed transfers.
- **Disaggregated Multi-Modal Execution:** Enables seamless communication between decoupled stages.
- **Multi-Instance Scaling:** Supports multiple instances for each omni stage, enabling elastic deployment and efficient load distribution across distributed clusters.



Async Chunk & Audio Streaming Output



E2E generation



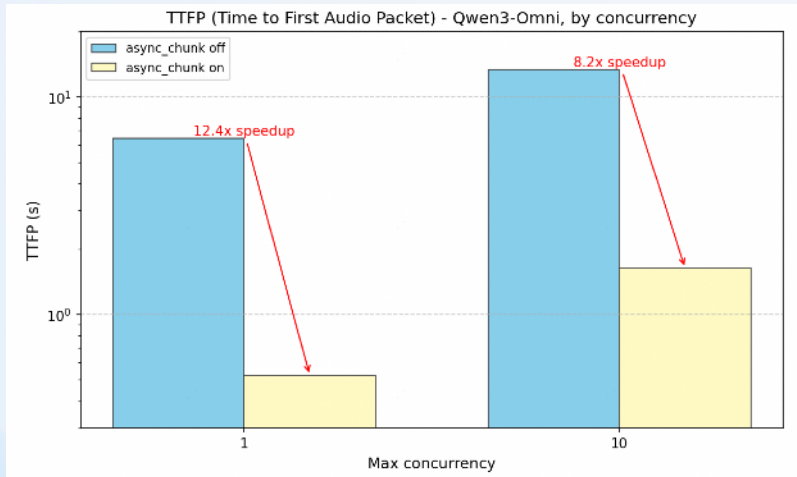
Streaming generation

- **Pipeline Between Stages:** Asynchronous chunked computation and communication across stages
- **Audio Streaming Output:** The waveform is output immediately after Talker generates each token.
- **API Server Support:** OpenAI v1/chat/completions with “stream” argument

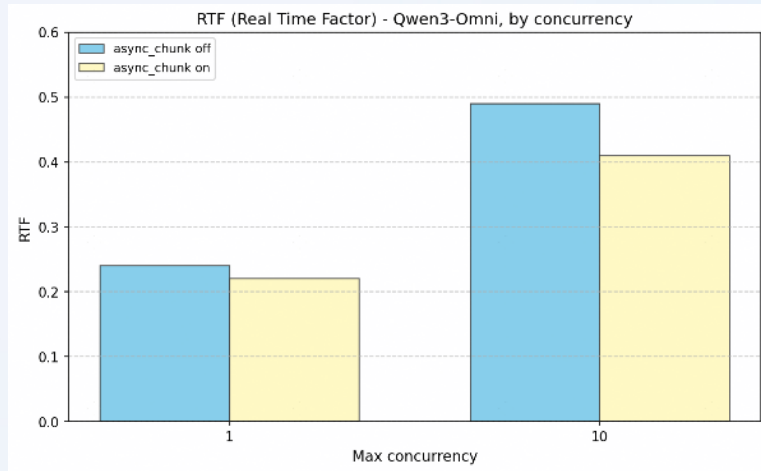


Async Chunk & Audio Streaming Output

- Qwen3-Omni



1 / 10 concurrency: 12.4x / 8.2x improves

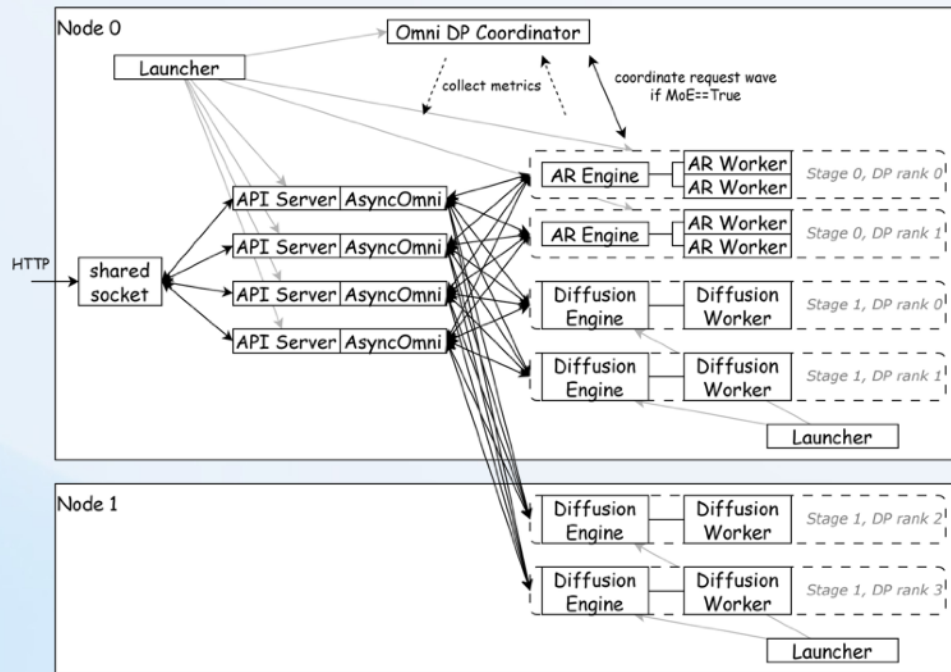


1 / 10 concurrency: 1.1 x / 1.2 x improves

https://docs.vllm.ai/projects/vllm-omni/en/latest/design/feature/async_chunk_design/

Scaling Disaggregated Serving

Example Multi-Node Deployment Diagram

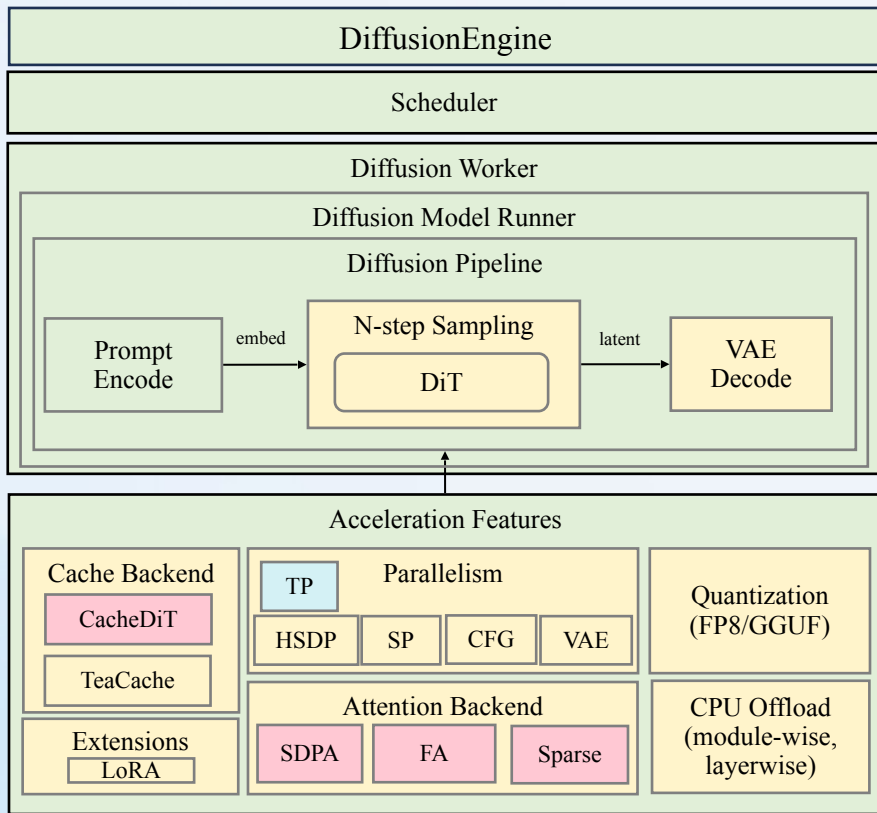


- **Command Line Interface (CLI) Design (PR #939)**
- **Distributed Deployment Structure**
 - a. Multiple APIServers (PR #939)
 - b. DP based Scaling for OmniStage
 - c. Coordinator for global management
- **Scheduling & Routing for load balancing**

Omni-Modality models Acceleration Support

Model	Output Modalities	Disaggregated deployment	Streaming	Async chunk	CUDA graph	Gradio demo
Qwen2.5-Omni	Text/Audio	✓	Only text	✗	✗	✓
Qwen3-Omni	Text/Audio	✓	✓	✓	✓	✓
MiMo-Audio	Text/Audio	✓	✓	✓	✓	✗
Qwen3-TTS	Audio	✓	✓	✓	✓	🕒
Bagel	Text/Image	✓	Only text	N/A	Only AR	✓
GLM-Image	Image	✓	N/A	N/A	✓	🕒

Diffusion Module Design



Legend: ■ vllm import ■ native ■ Third-party

Diffusion Core

- Natively implemented and optimized by configuring the acceleration layer
- Aligned with the AR arch with `model_runner` abstraction
- *Support step-wise scheduler with continuous batch for higher SLA*
- *Connect with AR module to speed up AR+DIT model inference with OmniConnector*

Acceleration Features

- Cache backend: Cache-DiT, TeaCache...
- Parallelism: TP/HSDP/SP/CFG/VAE...
- Attention: interface abstraction for third-party integration (FA/SAGE/MindIE-SD...)
- Quantization: FP8/GGUF...
- CPU Offload: module-wise/layer-wise
- Extensions: LoRA



When requests arrive

Request mm prompt:
"Make it play guitar"

generated image/
video



Omni
vllm_omni/entrypoints/
omni.py

OmniDiffusion
vllm_omni/diffusion/
omni_diffusion.py

- Convert user inputs into OmniDiffusionRequest
- Initialize DiffusionEngine

postprocess

DiffusionEngine
vllm_omni/diffusion/
diffusion_engine.py

- Preprocess
- Add to the scheduler's **waiting queue**

Scheduler
vllm_omni/diffusion/
schedule.py

DiffusionWorker
vllm_omni/diffusion/worker/
diffusion_worker.py

Worker.model_runner
vllm_omni/diffusion/worker/
diffusion_runner.py

ModelRunner.pipeline
vllm_omni/diffusion/models
pipeline_{model}_{task}.py

- **Waiting** request queue
- **Running** request queue
- **FIFO**



Diffusion modules with Acceleration Support

• X2I

• ⌚ = in progress (PR under review) • ✓ = supported • ✗ = not yet supported

Model	TeaCache	cache-DiT	SP (Ulysses & Ring)	CFG-Parallel	Tensor-Parallel	CPU Offload (Layerwise)	VAE-Patch-Parallel	FP8-Quantization
Bagel	✓	✓	✗	✗	✓	✗	✗	✗
FLUX.1-dev	⌚	✓	✗	✓	✓	✗	✗	✗
FLUX.2-klein	⌚	⌚	⌚	✓	✓	✗	✗	✗
GLM-Image	✗	⌚	✗	✗	✗	✗	✗	✗
LongCat-Image	✗	✓	✓	✓	✓	✗	✗	✗
LongCat-Image-Edit	✗	✓	✓	✓	✓	✗	✗	✗
Ovis-Image	✗	✓	✗	✓	⌚	✗	✗	✗
Qwen-Image	✓	✓	✓	✓	✓	✓	✗	✓
Qwen-Image-2512	✓	✓	✓	✓	✓	✓	✗	✓
Qwen-Image-Edit	✓	✓	✓	✓	✓	✓	✗	✗
Qwen-Image-Edit-2509	✓	✓	✓	✓	✓	✓	✗	✗
Qwen-Image-Layered	✗	✓	✓	✓	✓	✓	✗	✗
Stable-Diffusion3.5	⌚	✓	⌚	✓	✓	✗	✗	✗
Z-Image	✓	✓	✓	✗	✓ (TP=2 only)	✗	✓	✓

Summary

1. Support Omni-Modality model inference & serving
2. Consistent and unified API interface with vLLM
3. Native disaggregated deployment for different model stages
4. Pipeline async streaming for multiple stage engines
5. Native support for diffusion stage acceleration

Latest Release and Schedule



vLLM-Omni v0.16.0

 **Qwen3-Omni TTFL drops 90%**

121 Commits · ~60 Contributors (24 New )

-  **Ultra-Fast Audio & Video Engine**
Qwen3-Omni / TTS Performance Surge
TTFL  90% · RTF 0.22~45
-  **Production-Grade MiMo-Audio**
Speed up 11x faster · RTF ≈0.2
-  **Major Video & Image Generation (DIT) Upgrade**
 OPENAI-Compatible Video API
-  **DIT Extreme Acceleration & Quantization**
FP8/GGUF · TP/HSDP · SD 3.5

-  **New Models & Multimodal Expansion**
Video / Image: HunyuanImage3, OmniGen2, nextstep_1
-  **ComfyUI Integration**
-  **RDMA High-End Deployment**
Full Hardware: Intel, AMD, NPU

- **RC version**
Inherited new features from latest vLLM
Matched with vLLM formal release (e.g. v0.19.0rc1 released on Apr 04, 2026 for vLLM v0.19.0)
- **Formal version**
New features with sufficient tests
Every 4 weeks (e.g. v0.18.0 released on Mar 28, 2026)

vLLM-Omni Roadmap

- vLLM-Omni Team



Community collaboration

- Day-0 Support for xx
- Hardware support
- RL integration with veRL&veOmni

Future Roadmap (Overview)

<https://github.com/vllm-project/vllm-omni/issues/2136>

P0:

- CI/CD: Performance monitoring
- Streaming input and multi-turn
- Large scale deployment
- Full disaggregation(OmniConnector&OmniRouter...)
- RL Integration

P1:

- Wider model support(Omni&DiT&World model...)
- Diffusers backend support
- ModelRunnerV2 support
- Diffusion continuous batching

vLLM-Omni Networking Hour!



<https://github.com/vllm-project/vllm-omni>



<https://blog.vllm.ai/2025/11/30/vllm-omni.html>



<https://github.com/vllm-project/vllm-omni/blob/main/docs/assets/WeChat.jpg>

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全球软件开发大会

THANKS

大模型正在重新定义软件

Large Language Model Is Redefining The Software



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上海站

探索 AI 应用边界

Explore the limits of AI applications

2026 年 6 月 26-27 日 / 上海张江科学会堂



参会咨询

专题：世界模型与多模态智能突破

专题出品人：朱政

极佳视界 / 联合创始人 & 首席科学家



专题：Agent 系统架构与工程化实践

专题出品人：鲁洁楠

OPPO / 大模型算法负责人



专题：AI 原生数据工程

专题出品人：王彦辉

火山引擎 / 数智平台产品总监



专题：Agent 安全、评测与可信治理

专题出品人：马传雷（岳立）

支付宝 / 业务风险技术部负责人



专题：Agent 数据、记忆与运行时基础设施

专题出品人：邓亚峰

EverMind / CEO



专题：企业级研发体系的重构

专题出品人：沈浪

快手 / 研发效能负责人

